

4.6.3 Black body radiation (physics only)

4.6.3.1 Emission and absorption of infrared radiation

Content	Key opportunities for skills development
All bodies (objects), no matter what temperature, emit and absorb infrared radiation. The hotter the body, the more infrared radiation it radiates in a given time. A perfect black body is an object that absorbs all of the radiation incident on it. A black body does not reflect or transmit any radiation. Since a good absorber is also a good emitter, a perfect black body would be the best possible emitter.	

4.6.3.2 Perfect black bodies and radiation

Content	Key opportunities for skills development
Students should be able to explain: • that all bodies (objects) emit radiation • that the intensity and wavelength distribution of any emission depends on the temperature of the body. (HT only) A body at constant temperature is absorbing radiation at the same rate as it is emitting radiation. The temperature of a body increases when the body absorbs radiation faster than it emits radiation. (HT only) The temperature of the Earth depends on many factors including: the rates of absorption and emission of radiation, reflection of radiation into space. (HT only) Students should be able to explain how the temperature of a body is related to the balance between incoming radiation absorbed and radiation emitted, using everyday examples to illustrate this balance, and the example of the factors which determine the temperature of the Earth.	
(HT only) Students should be able to use information, or draw/interpret diagrams to show how radiation affects the temperature of the Earth's surface and atmosphere.	WS 1.2